

A Mathematical Theory of Communication, 1948

回看这篇论文 S1-S7:

1. "The fundamental problem of communication is that of reproducing at one point either exactly or approximately a message selected at another point."

通信的根本任务，是在一端精确或近似地复现另一端所挑选的消息。

2. "Frequently the messages have meaning; that is they refer to or are correlated according to some system with certain physical or conceptual entities."

消息背后的“语义 (Meaning)”存在，其关联着特定的物理实体或概念实体。同时，香农断言通信的语义层面与工程问题无关系，将工程问题的数据传输与哲学问题的含义理解相互彻底剥离。工程无关语义，信息量等价。

3. "The significant aspect is that the actual message is one selected from a set of possible messages. The system must be designed to operate for each possible selection, not just the one which will actually be chosen since this is unknown at the time of design."

工程上，信息转化是选择概率，通信系统要考虑“全集”和“不确定性”。香农的 Information Theory 是将 information 视为不确定性的消除。

4. "If the number of messages in the set is finite then this number or any monotonic function of this number can be regarded as a measure of the information produced when one message is chosen from the set, all choices being equally likely. As was pointed out by Hartley the most natural choice is the logarithmic function. Although this definition must be generalized considerably when we consider the influence of the statistics of the message and when we have a continuous range of messages, we will in all cases use an essentially logarithmic measure."

对信息的度量，信息量是所有的可能，自然数学选择对数函数作为基本工具（不仅是单调函数单调递增，而且有加法性质）。在这段，我们的假设是相等概率，后文我们会处理非等概率并引入香农熵。

5. "The choice of a logarithmic base corresponds to the choice of a unit for measuring information. If the base 2 is used the resulting units may be called binary digits, or more briefly bits, a word suggested by J. W. Tukey. A device with two stable positions, such as a relay or a flip-flop circuit, can store one bit of information. N such devices can store N bits, since the total number of possible states is 2^N and $\log_2 2^N = N$. If the base 10 is used the units may be called decimal digits. Since $\log_2 M = \frac{\log_{10} M}{\log_{10} 2} \approx 3.32 \log_{10} M$, a decimal digit is about $3\frac{1}{3}$ bits. A digit wheel on a desk computing machine has ten stable positions and therefore has a storage capacity of one decimal digit. In analytical work where integration and differentiation are involved the base e is sometimes

useful. The resulting units of information will be called natural units. Change from the base a to base b merely requires multiplication by $\log_b a$."

对数的底数是选择测量信息量的单位，介绍了“**Bit（比特）**”，比如说 N 个设备，总状态数是 2^N ，而 $\log_2 2^N = N$ 。如果底数是 10 的话，那就需要 $3\frac{1}{3}$ 比特。

6.

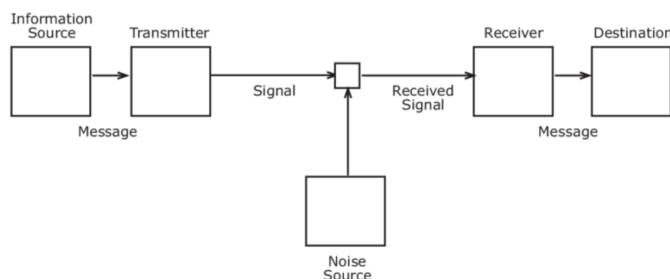


Figure 1: Schematic diagram of a general communication system

将图像/视频视为多维数学函数 $(f(x, y, t))$ ，发射机进行压缩、编码，接收机进行对称逆操作（重构）。值得注意的是，香农将“噪声源”视为一个独立作用于信道上的实体，这在我们研究判伪时有用。

7. "We wish to consider certain general problems involving communication systems. To do this it is first necessary to represent the various elements involved as mathematical entities, suitably idealized from their physical counterparts. We may roughly classify communication systems into three main categories: discrete, continuous and mixed. By a discrete system we will mean one in which both the message and the signal are a sequence of discrete symbols. A continuous system is one in which the message and signal are both treated as continuous functions, e.g. radio or television. A mixed system is one in which both discrete and continuous variables appear, e.g., PCM transmission of speech."

通信系统的通用方法论，就是将物理世界抽象为数学模型，将通信系统粗略地分为三大类：离散系统、连续系统和混合系统。比如，电报是离散系统，广播或电视是连续系统，语音的 PCM 传输是混合系统。香农当时将电视、广播作为连续系统，而当今除了极少数广播领域外，全球通信已经实现了 Discrete/PCM 范式。

8. "PART I: DISCRETE NOISELESS SYSTEMS, 1. THE DISCRETE NOISELESS CHANNEL: Generally, a discrete channel will mean a system whereby a sequence of choices from a finite set of elementary symbols $S_1 \dots S_n$ can be transmitted from one point to another. Each of the symbols S_i is assumed to have a certain duration in time t_i seconds (not necessarily the same for different S_i , for example the dots and dashes in telegraphy). It is not required that all possible sequences of the S_i be capable of transmission on the system; certain sequences only may be allowed. These will be possible signals for the channel. Thus in

telegraphy suppose the symbols are: (1) A dot, consisting of line closure for a unit of time and then line open for a unit of time; (2) A dash, consisting of three time units of closure and one unit open; (3) A letter space consisting of, say, three units of line open; (4) A word space of six units of line open. We might place the restriction on allowable sequences that no spaces follow each other (for if two letter spaces are adjacent, it is identical with a word space). The question we now consider is how one can measure the capacity of such a channel to transmit information."

离散信道的数学本质：从有限符号集中做离散选择的序列。假设每个符号 S_i 都有一定的持续时间 t_i 秒，对于不同的 S_i ，时间不一定相同。并不要求所有可能的 S_i 序列都能在该系统上传输，可能只允许某些特定序列，这些将是该信道允许的信号。而且，我们可以对允许的序列施加限制，约束会降低信道容量。然后，我们需要思考，请问如何度量这样一个信道传输信息的能力（信道容量）？

9. "In the teletype case where all symbols are of the same duration, and any sequence of the 32 symbols is allowed the answer is easy. Each symbol represents five bits of information. If the system transmits n symbols per second it is natural to say that the channel has a capacity of $5n$ bits per second. This does not mean that the teletype channel will always be transmitting information at this rate—this is the maximum possible rate and whether or not the actual rate reaches this maximum depends on the source of information which feeds the channel, as will appear later. In the more general case with different lengths of symbols and constraints on the allowed sequences, we make the following definition: The capacity C of a discrete channel is given by $C = \lim_{T \rightarrow \infty} \frac{\log_2 N(T)}{T}$. It is easily seen that in the teletype case this reduces to the previous result. It can be shown that the limit in question will exist as a finite number in most cases of interest. Suppose all sequences of the symbols $S_1, \dots, S_n$ are allowed and these symbols have durations $t_1, \dots, t_n$. What is the channel capacity? If $N(t)$ represents the number of sequences of duration t we have $N(t) = N(t - t_1) + N(t - t_2) + \dots + N(t - t_n)$. According to a well known result in finite differences, $N(t)$ is then asymptotic for large t to X_0^t where X_0 is the largest real solution of the characteristic equation: $X^{-t_1} + X^{-t_2} + \dots + X^{-t_n} = 1$ and therefore $C = \log_2 X_0$. In case there are restrictions on allowed sequences... In the telegraphy case mentioned above: $N(t) = N(t - 2) + N(t - 4) + N(t - 5) + N(t - 7) + N(t - 8) + N(t - 10) \dots$ Solving this we find $C = 0.539$."

以电传打字机的情况为例，有 32 个可用符号且允许任意序列，每个符号有 32 种同等可能性，那就是 5 比特信息 ($\log_2 32 = 5$)，如果系统每秒传输 n 个符号，那么该信道的容量为每秒 $5n$ 比特。但是这并不意味着该信道总是以这个速率传输信息的，实际上这是最大可能的速率，因为实际速率是否能够达到这个最大值，还要取决于供给信道的信源（比如文本是否有冗余等），“信道客观传输上限（信道容量）”与“实际传输速率”是不一样的。更何况，in the more general case, 我们还要考虑到符号长度不同且存在对允许序列的约束，因此我们需要注意到离散信道的容量 $C = \lim_{T \rightarrow \infty} \frac{\log_2 N(T)}{T}$ ，这是一个极限公式，表达的是上限。我们来看一下这个例子，首先他假设允许所有序列，然后问信道容量是多少？不难理解 $N(t) = N(t - t_1) + N(t - t_2) + \dots + N(t - t_n)$ ，根据齐次线性差

分方程的求解法则（类似于斐波那契数列 $F(n) = F(n-1) + F(n-2)$ 求解通项），我们假设 $N(t) = X^t$ （指数增长趋势）。代入方程得到：

$$X^t = X^{t-2} + X^{t-4} + X^{t-5} + X^{t-7} + X^{t-8} + X^{t-10}$$

两边同时除以 X^t ：

$$1 = X^{-2} + X^{-4} + X^{-5} + X^{-7} + X^{-8} + X^{-10}$$

为了方便计算，令变量换元 $\mu = X^{-1}$ （即 $X = \mu^{-1}$ ，或者说 $\mu = \frac{1}{X}$ ），方程就变成了：

$$1 = \mu^2 + \mu^4 + \mu^5 + \mu^7 + \mu^8 + \mu^{10}$$

这是一个 10 次多项式方程。利用数值计算方法，求得该方程唯一的介于 0 和 1 之间的正实根：

$$\mu_0 \approx 0.6874$$

因为 $X_0 = \frac{1}{\mu_0} = \frac{1}{0.6874} \approx 1.4547$ 。信道容量的定义是以 2 为底的对数 $\log_2 X_0$ （或者写成 $-\log_2 \mu_0$ ）：

$$C = \log_2(1.4547) = -\log_2(0.6874) \approx 0.539 \text{ bits/时间单位}$$

10. "A very general type of restriction which may be placed on allowed sequences is the following: We imagine a number of possible states $\alpha_1, \alpha_2, \dots, \alpha_m$. For each state only certain symbols from the set $S_1, \dots, S_n$ can be transmitted (different subsets for the different states). When one of these has been transmitted the state changes to a new state depending both on the old state and the particular symbol transmitted. The telegraph case is a simple example of this. There are two states depending on whether or not a space was the last symbol transmitted. If so then only a dot or a dash can be sent next and the state always changes. If not, any symbol can be transmitted and the state changes if a space is sent, otherwise it remains the same. The conditions can be indicated in a linear graph as shown in Fig. 2.

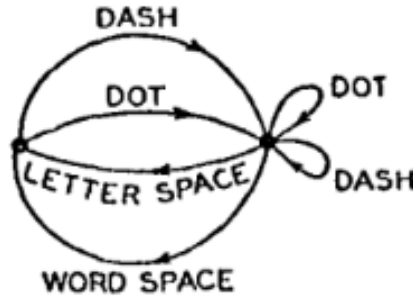


Figure 2: Graphical representation of the constraints on telegraph symbols

The junction points correspond to the states and the lines indicate the symbols possible in a state and the resulting state. In Appendix 1 it is shown that if the conditions on allowed sequences can be described in this form C will exist and can be calculated in accordance with the following result:

Theorem 1: Let $b_{ij}^{(s)}$ be the duration of the s -th symbol which is allowable in state i and leads to state j . Then the channel capacity C is equal to $\log W$ where W is the largest real root of the determinant equation: $\left| \sum_s W^{-b_{ij}^{(s)}} - \delta_{ij} \right| = 0$ where $\delta_{ij} = 1$ if $i = j$ and is zero otherwise. For example, in the telegraph case (Fig. 2) the determinant is:

$$\begin{vmatrix} -1 & (W^{-2} + W^{-4}) \\ (W^{-3} + W^{-6}) & (W^{-2} + W^{-4} - 1) \end{vmatrix} = 0$$

On expansion this leads to the equation given above for this case."

这边给予约束条件，仅能传输特定符号，当其中一个符号被传输后，状态会发生改变，转移取决旧状态及所传输的具体符号的新状态。还是以电报的情况为例，发不发空格，具体示例看看有向状态图。引出定理 1：

设 $b_{ij}^{(s)}$ 为在状态 i 中允许且会导致转移到状态 j 的第 s 个符号的持续时间。信道容量 C 等于 $\log W$ ，其中 W 是行列式方程的最大实根。其中，当 $i = j$ 时 $\delta_{ij} = 1$ ，否则为 0（即单位矩阵的元素）。然后我们就能得到上面电报案例的行列式表达式，这部分莫尔斯电码编码规则和 $C = 0.539$ 这个数值过时了，但是矩阵个例子本质是想说明和示范“受到规则约束的状态机 (Constrained State Machine)”，规则（有限状态机）限制了可能出现的信号组合，将约束形式化为“状态转移图（马尔可夫链/有限状态自动机）”。对后世有借鉴意义的是，在这里香农率先引入了状态机的思想（定理 1），证明了只要规则可以用有限状态机表达，就能通过线性代数的行列式（矩阵特征值）算出现实系统的极限容量。

11. "2. THE DISCRETE SOURCE OF INFORMATION: How is an information source to be described mathematically, and how much information in bits per second is produced in a given source? The main point at issue is the effect of statistical knowledge about the source in reducing the required capacity of the channel, by the use of proper encoding of the information."

从这开始研究信息本身（信源），关注的问题是**如何利用关于信源的统计知识，通过对信息进行恰当的编码，来降低对信道所需容量的要求。**

12. "We can think of a discrete source as generating the message, symbol by symbol. It will choose successive symbols according to certain probabilities depending, in general, on preceding choices as well as the particular symbols in question. A physical system, or a mathematical model of a system which produces such a sequence of symbols governed by a set of probabilities is known as a stochastic process. We may consider a discrete source, therefore, to be represented by a stochastic process. Conversely, any stochastic process which produces a discrete sequence of symbols chosen from a finite set may be considered a discrete source."

将离散信源想象为逐个符号地生成消息，且根据特定的概率选择后续符号，**Information 的 Statistical Structure**，转移概率 $P_i(j)$ ，人类语言、语音、电视信

号，本质上都可以被降维抽象为一串受概率支配的随机序列，只要掌握了信源的统计结构，就能用更短的代码来表示信息，数据压缩的本质是消除冗余 (Redundancy)。香农在此处建立的“信息源 = 概率分布 = 随机过程”这一范式，是当下 AI 大模型、无损压缩、计算机视觉和数字图像取证等共同遵循的 first principle。对于判伪具有的借鉴意义在于，鉴伪的根本，就是去检验这个随机过程的统计规律是否被动过手脚。

13. "3. THE SERIES OF APPROXIMATIONS TO ENGLISH"

A sufficiently complex stochastic process will give a satisfactory representation of a discrete source.

香农在这部分展示的逐阶近似与局部结构涌现。

14. "4. GRAPHICAL REPRESENTATION OF A MARKOFF PROCESS"

略。提到残余影响 (Residue of Influence)。所谓“上下文记忆”，本质上就是历史信息在当前时刻坍缩留下的“残余状态”。只要抓住了这个残余状态，就能以有限的数学结构（状态图）精准控制无限的符号生成。

15. "5. ERGODIC AND MIXED SOURCES"

遍历性，略。

16. "6. CHOICE, UNCERTAINTY AND ENTROPY:

Theorem 2: The only H satisfying the three above assumptions is of the form: $H = -K \sum_{i=1}^n p_i \log p_i$ where K is a positive constant.

We shall call $H = -\sum p_i \log p_i$ the entropy of the set of probabilities $p_1, \dots, p_n$." 创立“信息熵 (Information Entropy)”公式，香农通过连续性、单调性、可拆分加权性，推导了熵的表达形式：

$$H = -K \sum_{i=1}^n p_i \log p_i$$

熵是对“不确定性”的度量，也是消除这种不确定性所需要的最少平均信息量 (Bits)。信息的作用就是消除不确定性。

17. "7. THE ENTROPY OF AN INFORMATION SOURCE:

The entropy of the source will be defined as the average of these H_i , weighted in accordance with the probability of occurrence of the states in question: $H = \sum_i P_i H_i = -\sum_{i,j} P_i p_i(j) \log p_i(j)$

Theorem 3: Given any $\epsilon > 0$ and $\delta > 0$, we can find an N_0 such that the sequences of any length $N \geq N_0$ fall into two classes: 1. A set whose total probability is less than ϵ . 2. The remainder, all of whose members have probabilities satisfying the inequality $|\frac{\log p^{-1}}{N} - H| < \delta$.

Theorem 4: $\lim_{N \rightarrow \infty} \frac{\log n(q)}{N} = H$ when q does not equal 0 or 1."

略。Theorem 3 & 4，绝大多数序列发生的概率几乎为零，Typical Sequences 有 2^{HN} 个，且每一个的出现概率都接近 2^{-HN} 。意义：数据压缩不需要考虑所有 2^N 种情况，可以给 2^{HN} 个典型序列分配长度为 HN 比特的编码。